AI Exam Answer Grader

A Reusable NLP Asset for Rubric-Based Descriptive Answer Assessment

Weeks 3 & 4 Deliverable — Improved Approaches and Comparison | Target: 19 Sep – 2 Oct 2026

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1. Summary

In Weeks 1–2 we built a TF-IDF baseline (Approach A) that ran end-to-end but scored most answers far too low (MAE 2.63 on the Mohler pilot set). In Weeks 3–4 we cleaned the data, replaced the fixed similarity-to-marks formula with a mapping learned from human marks, moved to 5-fold cross-validation on unseen questions, and compared four improved NLP models against the baseline.

Our best model, a cross-encoder fine-tuned on the training answers, reduces the mean absolute error from 2.49 to 0.63 marks and raises the correlation with human marks from 0.47 to 0.59. It beats the calibrated baseline in 4 of 5 folds. It is packaged as Version 1 of the pipeline (evaluate_answer_v2) with the same JSON input/output contract as the baseline.

2. What Changed Since Weeks 1–2

Change	Why	Effect
Data cleaning	9 student answers were stored as "#NAME?", a spreadsheet error that destroyed the answer text but kept the human mark.	2,273 → 2,264 gradable answers
Score calibration	The Week 1–2 fixed linear scaling scored nearly every answer too low (mean human mark is 4.17 out of 5).	MAE 2.49 → 0.72 with the same TF-IDF features
5-fold cross-validation grouped by question	Our first Week 3–4 run used a single 80/20 split (16 test questions); the differences between methods were smaller than the variation between splits.	Every method is tested on unseen questions; results reported as mean ± standard deviation
Meaning-based NLP models	TF-IDF only matches exact words, so a correct answer in different words scores zero.	Two sentence-embedding models and a cross-encoder added
Fine-tuning	General semantic similarity is not the same as marking.	Cross-encoder learns to predict human marks directly; best MAE 0.63, r 0.59
Pipeline feedback and confidence	Week 1–2 feedback could contradict the score, and confidence was almost constant (≈0.99).	Feedback flags paraphrases and wrong explanations; confidence now varies with model agreement

3. Dataset and Experimental Setup

Dataset: Mohler short-answer grading dataset (Kaggle mirror abdokamr/mohler): answers to data-structures questions from an undergraduate computer science course, each marked 0–5 by two human graders. We use the average of the two graders (score_avg) as the label. After cleaning: 2,264 answers to 79 unique questions (two questions share identical text). The mean mark is 4.17, so the data is skewed towards correct answers; for this reason a mean predictor is included as a floor that any real method must beat.

Split: the 79 questions are shuffled and divided into 5 groups of 15–16 questions (423–487 answers each). All answers to one question stay in the same group. Each method is trained on 4 groups and tested on the fifth, repeated 5 times, so every result is measured on questions the model has never seen — the real use case, where faculty set a new exam question.

Metrics: mean absolute error (MAE), root mean squared error (RMSE), agreement within ±1 mark, and Pearson correlation with human marks, as required for grading tasks in the project handout.

4. Methods Compared

Label	Method
Floor	Mean predictor: always predicts the average training mark.
A	Week 1–2 baseline: TF-IDF cosine similarity + keyword concept coverage, fixed linear scaling to marks.
A (calibrated)	Same TF-IDF features, mapped to marks by a linear regression learned on the training questions.
B1 MiniLM	Sentence-embedding similarity (all-MiniLM-L6-v2) + concept coverage, calibrated.
B1 MPNet	Larger sentence-embedding model (all-mpnet-base-v2) + concept coverage, calibrated.
B3 pre-trained	Cross-encoder (cross-encoder/stsb-TinyBERT-L4), which reads the reference and student answer together; its similarity score + concept coverage, calibrated.
B3 fine-tuned	The same cross-encoder fine-tuned in each fold on the training answers to predict mark ÷ 5 (3 epochs, batch size 16, learning rate 4e-5, binary cross-entropy loss).

Label B2 is reserved for the LLM rubric-grounded grader planned for Week 5.

5. Results — Baseline vs Improved

Mean ± standard deviation over 5 folds, on unseen questions. The best model overall (lowest MAE and highest correlation) is highlighted.

Method	MAE ↓	RMSE ↓	Within ±1 (%) ↑	Pearson r ↑
Floor: Mean predictor	0.909 ± 0.036	1.115 ± 0.093	79.9 ± 5.5	–
A: TF-IDF (Week 1–2)	2.486 ± 0.204	2.795 ± 0.182	15.0 ± 5.6	0.467 ± 0.067
A: TF-IDF (calibrated)	0.724 ± 0.087	0.975 ± 0.095	72.4 ± 5.0	0.486 ± 0.071
B1: MiniLM (calibrated)	0.700 ± 0.066	0.936 ± 0.075	74.3 ± 2.2	0.543 ± 0.043
B1: MPNet (calibrated)	0.701 ± 0.072	0.938 ± 0.081	74.4 ± 3.2	0.537 ± 0.053
B3: Cross-encoder pre-trained (calibrated)	0.702 ± 0.064	0.944 ± 0.076	75.7 ± 4.0	0.531 ± 0.063
B3: Cross-encoder fine-tuned	**0.634 ± 0.108**	**0.923 ± 0.127**	**75.5 ± 5.7**	**0.589 ± 0.099**

Pearson correlation is undefined for the mean predictor because its predictions are constant. The Week 1–2 baseline scored MAE 2.63 in its original single run; 2.49 is the same method under the new cross-validation setup.

Is the improvement real?

Number of folds (out of 5) in which each improved method has a lower MAE than calibrated TF-IDF, and the average MAE gain:

- B1 MiniLM: 4 of 5 folds, average gain 0.024
- B1 MPNet: 5 of 5 folds, average gain 0.022
- B3 cross-encoder pre-trained: 4 of 5 folds, average gain 0.022
- **B3 cross-encoder fine-tuned: 4 of 5 folds, average gain 0.090**

6. Key Findings

1. **Calibration was the largest single improvement.** With the same TF-IDF features, learning the similarity-to-marks mapping from human marks cut MAE from 2.49 to 0.72. The main weakness of the Week 1–2 baseline was its scale, not its features.

2. **Meaning-based models give a smaller but consistent gain.** All three pre-trained neural models beat calibrated TF-IDF in at least 4 of 5 folds. The gain shows mainly in correlation (0.53–0.54 vs 0.49), i.e. better ranking of good answers above weak ones.
3. **Fine-tuning gives the best result.** The fine-tuned cross-encoder has the lowest MAE (0.634) and the highest correlation (0.589), because it learns what the human graders reward rather than general sentence similarity.
4. **A bigger embedding model did not help.** MPNet performed the same as the smaller MiniLM on this task.
5. **Limitations of the best model.** It varies the most between folds (± 0.108), it still makes occasional large errors (RMSE improves less than MAE), and the mean predictor remains best on ±1 agreement (79.9%) because most answers in Mohler score 4–5.
6. **The Week 1–2 paraphrase failure is not solved.** On the 168 answers with zero TF-IDF word overlap but a human mark of 4 or more, calibrated TF-IDF has MAE 1.12 and the neural models 1.17–1.22. Selecting only high-scoring answers favours any model that predicts near the average, so this subset alone does not rank the methods fairly; the conclusion is that no current method reliably recognises short correct answers written very differently from a short reference.

7. Error Analysis (Best Model)

Representative large errors of the fine-tuned cross-encoder, from the cross-validation predictions:

Question (reference)	Student answer	Human	Model	Failure type
Steps to search a node in a BST ("The height of the tree.")	log n	4.75	1.13	Correct, but needs domain knowledge
Advantage of linked lists over arrays ("dynamic structures...")	There is no predetermined length	5.00	1.45	Correct paraphrase, no shared words
What is a leaf? ("A node that has no children.")	The elements at the very bottom of an inverted tree ... are called leaves.	5.00	1.57	Correct paraphrase, longer wording
What is a queue? ("...first in first out...")	like a linked list expect first in last out	1.25	4.70	Wrong: one word reverses the meaning
Where do C++ programs begin to execute? ("At the main function.")	in the testing phase	0.00	3.49	Wrong answer on a related topic
Comparisons to find an element in a BST ("The height of the tree...")	(n^k)-1 where k is the height of the tree	1.50	4.94	Right words, wrong answer

The errors fall into two types. Correct answers that need domain knowledge or use very different wording are under-scored, and wrong answers that share the reference's topic or vocabulary are over-scored. Both require judging whether an answer is correct, not just whether it is similar. This is the main motivation for Approach B2 in Week 5, where an LLM checks each rubric concept individually before assigning a score.

8. Version 1 Pipeline — evaluate_answer_v2()

The reusable function keeps the Week 1–2 contract: it takes question, reference_answer, student_answer, maximum_marks and question_id, and returns a structured JSON object. It will be wrapped as POST /evaluate-answer in Week 6.

- **Score:** fine-tuned cross-encoder trained on all 2,264 cleaned answers, scaled to the question's maximum marks and rounded to 0.5.
- **Confidence:** agreement between two independent models (cross-encoder and calibrated MPNet): 1 − |difference| ÷ 5. Low agreement means the answer should be checked by a human.
- **Feedback:** key reference terms covered and missing, plus a note when the score and the keyword match disagree (correct meaning in different words, or right terms with a wrong explanation).

Example output for a paraphrased correct answer (reference: "Through iteration."; student: "You can rewrite it as a loop, using an iterative solution with an explicit stack."):

```
{ "question": "Q_paraphrase", "score": 4.5, "maximum_score": 5,
  "feedback": "Covers none of the key reference terms. Missing: iteration. The answer
   appears correct in meaning but uses different wording from the reference.",
  "confidence": 0.73 }
```

The demo examples are a format demonstration only; performance claims are based on the cross-validation results in Section 5.

9. Responsible AI Notes

- **Advisory only:** the grader recommends a score; final marks remain a faculty decision.
- **Privacy:** all models in this stage run locally, so no answer text is sent to an external service. Mohler is a public dataset.
- **Known failure types:** the two error types in Section 7 are documented so that low-confidence and disputed answers can be routed to human review.
- **Bias and transfer:** the models are trained on one computer science course marked by two graders, and may not transfer to other subjects or writing styles, such as answers with spelling errors or non-native English phrasing.

10. Next Steps (Week 5)

- **Approach B2 — LLM rubric-grounded grader:** a two-step design (build a rubric of key concepts per question, then mark each concept as present, partial or missing before scoring) has been prototyped; evaluation on the same 5 folds is pending API access.
- **Full error analysis:** categorise the largest errors of every method, not only the best one.
- **Class dataset:** build a labelled test set from our own anonymized mid-term answers, with rubrics prepared with faculty.
- **Confidence estimation:** check whether low-confidence predictions really are the less accurate ones (planned with the Week 6 API work).

Code and results: Group2_AI_Exam_Grader_Week3-4.ipynb; results/cv_comparison_table.csv, results/cv_fold_results.csv, results/out_of_fold_predictions.csv.